

Simple Approaches to Inference with Difference-in-Differences Estimators with Small Cross-Sectional Sample Sizes

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Abstract

We propose simple methods for valid inference in panel data difference-in-differences (DiD) settings with a small number of treated or control units, including cases in which both groups are small. The approach uses a suitable transformation to collapse the panel into a cross-sectional data set. If the classical linear model assumptions hold in the resulting cross-sectional regression, exact inference is available, even with only two control units and one treated unit. The approach is expected to work best with many pre- and post-intervention periods, so that the central limit theorem over time makes normality a better approximation. Nevertheless, under joint normality and homoskedasticity of the time-varying errors, exact inference can also be applied with few time periods and few cross-sectional units. We also consider randomization inference and HC3 heteroskedasticity-robust inference as alternatives to exact inference. The approach can additionally accommodate unit-specific trends and, with sufficiently large samples, control variables. We illustrate the approach using California's 1989 cigarette tax increase in a single-treated-unit application and the expansion of "castle" laws in the United States under staggered intervention timing.

Keywords: Difference-in-differences; Small N; Synthetic Control; Synthetic DiD

1 Introduction

Difference-in-differences methods with panel data have a long history for evaluating policy interventions. In the case of common intervention timing, various methods are available for allowing treatment effects to vary by treatment period, as well as by control variables. Wooldridge (2025b) shows how to estimate flexible linear models by pooled OLS or fixed effects – they are equivalent – that allows straightforward inference, provided there are enough cross-sectional units and the data are independently sampled across units. With a relatively small time series sample size (T) and large enough numbers of control (N_0) and treated units (N_1), standard errors robust to arbitrary serial correlation and heteroskedasticity are readily available. Other methods – such as the Callaway and Sant’Anna (2021) long differencing methods – also rely on large cross-sectional sample sizes for accurate inference.

As shown in Lee and Wooldridge (2026) [LW (2026)], regression-based difference-in-differences (DID) estimators, either with common timing or staggered interventions, can be obtained via cross-sectional regressions after simple time-series transformations of the data at the unit level. This characterization of DiD estimators permits application of many widely used treatment effects estimators in the DiD setting, including matching, doubly robust estimators using the propensity score, and even machine learning causal methods. Here we use the representations in LW (2026) to obtain simple inference when the usual inference obtained from clustering at the individual unit may be problematical.

Our approach in the current paper combines the algebraic equivalence in LW (2026) and an idea from Donald and Lang (2007), who propose collapsing individual-level data to aggregated groups when the assignment of an intervention is at the group level. Here, the collapsing of the data is at the unit level across time in order to exploit the equivalences of panel data DiD estimators and those based on cross-sectional regressions. In addition to the standard DiD estimators, the approach extends easily to allow for unit-specific trends. Also, time-constant control variables can be added if the cross-sectional sample sizes are large enough to avoid degeneracies.

The method can also be useful in cases where N_0 and N_1 (the sizes of the control and treatment groups, respectively) are not particularly small but where the number of time

series is reasonably large. In such cases, using the panel structure and clustering by cross-sectional unit to account for serial dependence can cause distortions in the inference. The approach we suggest here requires no modification for large T . In fact, because inference is conducted on the transformed cross section, we do not need to model the within-unit serial dependence directly, as long as the transformed errors satisfy the required assumptions.

Recently, [Simonsohn \(2021\)](#), in commenting on [Young \(2019\)](#), argues that a particular version of the heteroskedasticity-robust standard error, proposed by [MacKinnon and White \(1985\)](#) (and labeled “HC3” in the popular Stata statistical package), can produce satisfactory results even in Young’s setting with $N_0 = 18$ and $N_1 = 2$. Therefore, in some cases one might feel comfortable using heteroskedasticity-robust inference.

A popular method for studying interventions with a small number of treated units (with one being the most common) is the synthetic control method (SCM), pioneered by [Abadie, Diamond and Hainmueller \(2010\)](#) [[ADH \(2010\)](#)], where pre-intervention data are used to obtain a weighted average of “donor” control units to obtain a single synthetic control. More recently, [Arkhangelsky, Athey, Hirshberg, Imbens and Wager \(2021\)](#) [[AAHIW \(2021\)](#)] propose a unification of traditional DiD and SC methods, called *Synthetic difference-in-differences* (SDiD). The SDiD approach allows more flexibility than either DiD or SC, and inference methods are available. However, SDiD assume that the series are weakly dependent – often called “integrated of order zero,” or $I(0)$ – and that N_0 , T_0 , and T_1 are suitably “large.” The authors also impose normality in obtaining inference, although it is not clear how important that is in practice.

Compared with SDiD, the method we propose in this paper need not require a very large control group or a large number of time periods. Because inference is based on a cross-sectional regression under the assumptions of the classical linear model, there are minimal restrictions on N_0 , N_1 , T_0 , and T_1 . Moreover, we can use a cross-sectional regression for each treated time period, allowing for separate estimated effects and confidence intervals over time. We can also aggregate these estimates to obtain a single effect. Our framework additionally accommodates heterogeneous untreated outcome patterns, such as unit-specific linear trends and seasonality.

These differences imply a tradeoff rather than a clear ranking of the methods. SC and SDiD use the donor pool and pre-treatment outcomes to construct data-driven coun-

terfactuals, whereas our approach imposes more explicit structure on untreated outcome evolution. When that structure is a good approximation, the transformation-based approach can perform well with either small or large donor pools, while remaining applicable even in very small cross-sectional samples. We therefore view the methods proposed here as complementary to SC and SDiD.

Section 2 considers the common-timing case and develops cross-sectional estimation and inference after transforming the panel data. Section 3 extends the framework to accommodate unit-specific trends, seasonal effects, and aggregated outcomes. In Section 4, we compare our approach with synthetic control (SC) and synthetic difference-in-differences (SDiD), with particular attention to the different structures used to construct the counterfactual and conduct inference.

Section 5 presents Monte Carlo simulations comparing alternative inference procedures when the number of treated or control units is small. In Section 6, we revisit California’s 1989 cigarette tax increase using the data in [Abadie, Diamond and Hainmueller \(2010\)](#). Section 7 extends the approach to staggered intervention timing and applies it to the expansion of “castle” laws and homicide rates. Section 8 discusses additional practical considerations, including the choice of pre-treatment periods and inference with clustering or spatial correlation in larger cross-sectional samples. We conclude in Section 9.

2 The Common Timing Case

We begin with the common-timing setting, in which a total of T time periods are observed and an intervention occurs at time $S \in \{2, \dots, T\}$. Each cross-sectional unit i is either treated from period S onward or remains untreated throughout the sample. Let

$$D_i = \begin{cases} 1, & \text{if unit } i \text{ is treated,} \\ 0, & \text{if unit } i \text{ is a control unit.} \end{cases}$$

We define the time-varying treatment indicator as

$$W_{it} = D_i \cdot post_t,$$

where $post_t = 1\{t \geq S\}$. We assume independence across cross-sectional units. Within a unit, the outcome process may exhibit serial dependence and time-varying distributions

across t . Later, we provide a discussion of how one can accommodate clustering or spatial correlation if the cross-sectional sample sizes are reasonably large.

The simple DiD estimator can be obtained as the coefficient $\hat{\tau}_{DD}$ on W_{it} from the pooled regression

$$Y_{it} \text{ on } 1, D_i, post_t, W_{it}, \quad i = 1, \dots, N; \quad t = 1, \dots, T \quad (2.1)$$

Equivalently, one can replace D_i and $post_t$ with a full set of unit and time period fixed effects, resulting in the popular two-way fixed effects (TWFE) estimator. [The equivalence follows from Wooldridge (2025b, 2010).]

When N is small (or N_1 is small), inference using the TWFE estimator is tricky because one should allow for serial correlation in the underlying time-varying errors, U_{it} , that are implicit in the equation. The usual method of clustering by unit i generally results in poor performance with small N (or with small N_0 or N_1), which motivates the alternative inference methods developed in this paper.

2.1 Identification and Estimation of a Single ATT

We begin by considering a single ATT that averages treatment effects over the post-treatment periods. We first present the usual DiD estimator and then show its equivalence to our unit-specific demeaning procedure, motivated by Lee and Wooldridge (2026). We then establish the conditions under which this common estimator identifies the ATT.

For each unit i , create

$$\begin{aligned} \Delta \bar{Y}_i &\equiv \frac{1}{T - S + 1} \sum_{t=S}^T Y_{it} - \frac{1}{S - 1} \sum_{t=1}^{S-1} Y_{it} \\ &\equiv \bar{Y}_{i,post} - \bar{Y}_{i,pre}. \end{aligned} \quad (2.2)$$

which is a simple transformation of the data within each unit. This results in a cross-sectional data set $\{(\Delta \bar{Y}_i, D_i) : i = 1, \dots, N\}$, where N can be large or small. In the second step, $\hat{\tau}_{DD}$ is obtained as the coefficient on D_i from the simple regression

$$\Delta \bar{Y}_i \text{ on } 1, D_i, \quad i = 1, \dots, N. \quad (2.3)$$

Of course, this regression leads to the formula for a difference in means:

$$\begin{aligned} \hat{\tau}_{DD} &= N_1^{-1} \sum_{i=1}^N D_i \cdot \Delta \bar{Y}_i - N_0^{-1} \sum_{i=1}^N (1 - D_i) \cdot \Delta \bar{Y}_i \\ &= \Delta \bar{Y}_{treat} - \Delta \bar{Y}_{control}. \end{aligned} \quad (2.4)$$

The preceding representation is simply the usual DiD estimator expressed as a cross-sectional difference in average outcome changes. Motivated by the transformation approach of [Lee and Wooldridge \(2026\)](#), we next introduce the unit-specific demeaning procedure used in this paper. Each unit’s pre-treatment mean serves as a benchmark for its post-treatment outcomes, which are then aggregated to obtain a single transformed outcome for cross-sectional estimation and inference.

Procedure 2.1 (Unit-Specific Demeaning):

1. Obtain $\bar{Y}_{i,pre}$, for each i , using the pre-intervention regression:

$$Y_{it} \text{ on } 1, \quad t = 1, \dots, S-1, \quad (2.5)$$

where the coefficient (the only coefficient) gives $\bar{Y}_{i,pre}$.

2. Following [Lee and Wooldridge \(2026\)](#), in each post-intervention period, obtain out-of-sample residuals, or prediction errors:

$$\dot{Y}_{it} = Y_{it} - \bar{Y}_{i,pre} = Y_{it} - \frac{1}{S-1} \sum_{r=1}^{S-1} Y_{ir}, \quad t = S, \dots, T. \quad (2.6)$$

3. For the single-effect setting, average the out-of-sample residuals over the post-intervention periods to obtain the same regressand as in (2.3):

$$\bar{\dot{Y}}_i \equiv \frac{1}{T-S+1} \sum_{t=S}^T \dot{Y}_{it} = \bar{Y}_{i,post} - \bar{Y}_{i,pre} = \Delta \bar{Y}_i.$$

4. Obtain the average effect, $\hat{\tau}_{DM}$, and its standard error and confidence interval from

$$\bar{\dot{Y}}_i \text{ on } 1, D_i, \quad i = 1, \dots, N. \quad (2.7)$$

Thus, our proposed Procedure 2.1 produces the same estimate as the simple DiD regression. Accordingly, both estimators identify the ATT under the same no-anticipation (NA) and parallel-trends (PT) assumptions. We follow [Lee and Wooldridge \(2026\)](#) in stating these identification conditions.

With $Y_{it}(0)$ and $Y_{it}(1)$ denoting the potential outcomes for unit i in period t in the absence of treatment and under treatment, respectively, a weak form of the NA assumption is

$$E[Y_{it}(1) - Y_{it}(0) | D_i = 1] = 0, \quad t = 1, \dots, S-1, \quad (2.8)$$

so the potential outcomes are the same, on average for the treated units. Sufficient, of course, is $Y_{it}(1) = Y_{it}(0)$, $t = 1, \dots, S - 1$.

The parallel trends assumption is that, for all $t = 2, \dots, T$,

$$E[Y_{it}(0) - Y_{i1}(0) | D_i] = E[Y_{it}(0) - Y_{i1}(0)] \equiv \delta_t, \quad (2.9)$$

where δ_t is a constant that is unrestricted over time. This assumption allows an unrestricted trend in the control state, but that trend must be the same across control and treated units. In effect, it allows treatment assignment, D_i , to be correlated with the level, say $Y_{i1}(0)$, but it rules out cases where the assignment is based on differential trends in the control state.

As shown in [Lee and Wooldridge \(2026\)](#), the PT assumption implies that, for some constant α ,

$$E[\Delta \bar{Y}_i(0) | D_i] = \alpha, \quad (2.10)$$

where

$$\Delta \bar{Y}_i(0) \equiv \bar{Y}_{i,post}(0) - \bar{Y}_{i,pre}(0)$$

is the same transformation as in (2.2), but applied to the untreated potential outcomes. Thus, $\Delta \bar{Y}_i(0)$ is mean independent of treatment status. Together with the NA assumption, (2.10) implies that τ is identified and that $\hat{\tau}_{DD}$ is conditionally unbiased and consistent; see [Lee and Wooldridge \(2026\)](#). The unbiasedness result is particularly important here because we allow either N_1 or N_0 to be small.

Equivalently, under NA and PT,

$$E[\Delta \bar{Y}_i | D_i] = \alpha + \tau D_i. \quad (2.11)$$

The benefit of this characterization is that we can think of it as a simple underlying model,

$$\Delta \bar{Y}_i = \bar{Y}_i = \alpha + \tau D_i + U_i, \quad (2.12)$$

with

$$E(U_i | D_i) = 0, \quad (2.13)$$

where the zero conditional mean condition follows from the DiD identification assumptions for the ATT, τ . This representation provides the basis for the finite-sample inference procedures considered below.

2.2 Exact Inference with Small Cross-Sectional Samples

The cross-sectional representation in (2.12) provides a particularly simple framework for inference. If N is reasonably large, and we have several treated and control units, we can rely on asymptotic analysis and justify a heteroskedasticity-robust standard error for $\hat{\tau}_{DD}$ for computing a confidence interval for τ . But when N is small, or, say, N_0 is large but N_1 is small, we cannot rely on asymptotics. Nevertheless, it is possible that (2.12) satisfies the classical linear model (CLM) assumptions:

$$U_i \mid D_i \sim \text{Normal}(0, \sigma_U^2). \quad (2.14)$$

Under (2.12) and (2.14), conditional on D_i , $\hat{\tau}_{DD}$ has an exact normal distribution, and

$$\frac{\hat{\tau}_{DD} - \tau}{\text{se}(\hat{\tau}_{DD})} \sim \mathcal{T}_{N-2}. \quad (2.15)$$

This result means that we can obtain exact tests of any null hypothesis, such as $H_0 : \tau = 0$. Moreover, the usual confidence intervals, obtained using percentiles of the $\mathcal{T}_{N-2}$ distribution, have exact coverage.

Relatedly, Hagemann (2025) assumes normality in the context of testing for a treatment effect when a single cluster is treated with many units within the cluster, allowing for different variances in the distributions of the control and treated units. Here, we do not require any particular number of time periods (units within a cluster), provided the normality assumption in (2.14) holds.

Looking at the expression for $\hat{\tau}_{DD}$ in (2.4), it is clear why having a large overall cross-sectional sample is not sufficient for asymptotic inference. For example, even if N_0 is very large, asymptotic theory will not generally provide a good approximation if N_1 remains small. We need the central limit theorem to apply to both terms in (2.4), and the same issue arises when N_0 is small.

Assuming (2.14) holds, the exact approach works in any setting with $N_0 \geq 1$, $N_1 \geq 1$, and $N = N_0 + N_1 \geq 3$. In particular, we can have a single treated unit, $N_1 = 1$, and many or few control units. When there is only a single treated unit, the t -statistic

$$\hat{\tau}_{DD}/\text{se}(\hat{\tau}_{DD})$$

is known as the “studentized residual,” which plays a role in outlier analysis. See, for example, Wooldridge (2025a, Section 9.5). Viewing the t -statistic for $\hat{\tau}_{DD}$ as an outlier

diagnostic makes intuitive sense: we are trying to determine whether the single treated unit, with $D_i = 1$, is an “outlier” compared with the control units.

The approach just outlined is essentially that taken by [Donald and Lang \(2007\)](#), but here we apply it to panel data with either a short or long stretch of time. If pre- and post-intervention periods (T_0 and T_1) are reasonably large, the normality assumption in (2.14) may be an acceptable assumption because, if the data are weakly dependent across time, the central limit theorem implies that $\bar{Y}_{i,post}$ and $\bar{Y}_{i,pre}$ are approximately normal. Moreover, because we require conditional normality only of

$$\bar{\dot{Y}}_i = \bar{Y}_{i,post} - \bar{Y}_{i,pre}$$

strong dependence, such as a time-constant unobserved effect, is eliminated by differencing the post-intervention and pre-intervention averages. Underlying unit root processes also need not cause $\bar{\dot{Y}}_i$ (equivalently, $\Delta\bar{Y}_i$) to deviate substantially from normality.

2.3 Alternative Inference: HC3 and Randomization Inference

An advantage of our proposed method is that it separates estimation from the choice of inference procedure. Once the data have been reduced to one transformed outcome per unit, the same point estimator can be paired with different inferential methods depending on the assumptions one is willing to impose. In addition to the exact t procedure described above, we consider HC3-based inference and randomization inference (RI).

The exact t procedure relies on homoskedasticity in addition to conditional normality. When heteroskedasticity is a concern and the cross-sectional sample is not extremely small, a heteroskedasticity-robust covariance estimator provides a natural alternative. We focus on the HC3 adjustment of [MacKinnon and White \(1985\)](#).

Simulation evidence from [Simonsohn \(2021\)](#) is encouraging in showing that HC3 can perform well even with a relatively small number of treated and control units. Thus, heteroskedasticity-robust inference may provide a useful alternative except in settings with a very small number of treated and/or control units. At the same time, HC3 does not make the finite-sample t distribution exact, and it should not be interpreted as repairing violations of normality or cross-unit independence. Rather, it provides a heteroskedasticity-robust approximation that may be useful when there are enough treated and control units

A second inferential option available within our framework is randomization inference

(RI), which does not rely on a normal model for the transformed errors. RI is appropriate when treatment status can reasonably be viewed as exchangeable across the cross-sectional units under the null hypothesis. In our implementation, treatment status is randomly re-assigned across units while holding the number of treated units fixed, and the observed treatment effect is compared with the distribution of effects obtained from these reassignments. The RI p -value measures how unusual the observed effect is relative to this randomization distribution.

Table 1 summarizes inference options available for our proposed transformation-based estimator.

Table 1: Inference options for the proposed transformation-based estimator

Method	Main requirement	Inference type
Exact OLS t	Cross-unit independence; normality and homoskedasticity of transformed errors	Finite-sample exact
HC3	Cross-unit independence; heteroskedasticity allowed	Heteroskedasticity-robust
Randomization inference	Treatment can be plausibly reassigned across units	Permutation-based

To facilitate implementation of the proposed approach, we provide a publicly available user-written Stata command, `lwdid`. By default, the command reports the usual t -based inference for the transformed cross-sectional regression. HC3 standard errors and randomization-inference p -values can be obtained through additional options. Thus, the same transformation-based point estimator can be paired with different inference procedures depending on the assumptions deemed appropriate for the application.

2.4 Adding Control Variables

The preceding analysis extends naturally to settings with observed control variables. As shown in LW (2026), the PT assumption is easily relaxed by conditioning on controls. Given time-constant controls $\mathbf{X}_i$, a $1 \times K$ vector, the conditional PT assumption is

$$E[Y_{it}(0) - Y_{i1}(0) | D_i, \mathbf{X}_i] = E[Y_{it}(0) - Y_{i1}(0) | \mathbf{X}_i] \equiv \alpha_t + \mathbf{X}_i \beta_t, \quad (2.16)$$

where we have imposed linearity because we have little choice in a small- N setting. Assumption (2.16) is an unconfoundedness assumption on D_i once we condition on $\mathbf{X}_i$, but

it is in terms of the differences $Y_{it}(0) - Y_{i1}(0)$, rather than the levels $Y_{it}(0)$. (This is what gives the DiD procedure its main advantage over cross-sectional regression.) In terms of the cross-sectional regression, it is easy to adapt LW (2026) to see that we should add $\mathbf{X}_i$ to the equation. To use exact inference, we would write

$$\Delta \bar{Y}_i = \alpha + \tau D_i + \mathbf{X}_i \beta + U_i \quad (2.17)$$

$$U_i | D_i, \mathbf{X}_i \sim \text{Normal}(0, \sigma_U^2) \quad (2.18)$$

Provided $N > K+2$, (2.17) can be estimated by OLS, and, under the conditional normality and homoskedasticity assumptions, the t statistic has an exact $\mathcal{T}_{N-K-2}$ distribution and the corresponding confidence intervals are exact. Again, it is intriguing to think that, even without N being too large, we might use heteroskedasticity-robust inference using HC3.

When N_0 and N_1 are both sufficiently large, and the support of $\mathbf{X}$ for $D = 1$ is contained in that for $D = 0$, full regression is preferred on theoretical grounds. Namely, $\hat{\tau}_{DD}$ is the coefficient on D_i from the regression that includes interactions:

$$\Delta \bar{Y}_i \text{ on } 1, D_i, \mathbf{X}_i, D_i \cdot (\mathbf{X}_i - \bar{\mathbf{X}}_1), \quad i = 1, \dots, N,$$

which is identical to running separate regressions for $D_i = 0$ and $D_i = 1$. Here, $\bar{\mathbf{X}}_1$ is the average of the covariates over $D_i = 1$. Unfortunately, this cannot be done with small N_1 , as it requires $N_0 > K+1$ and $N_1 > K+1$.

2.5 Estimating Separate Effects for Each Treated Period

Rather than estimate a single ATE, it is little additional effort to estimate ATEs separately in each treated period. Define $\dot{Y}_{it}$ as in (2.6), and this is used as the dependent variable in period t . Then $\hat{\tau}_{t,DD}$ for $t = S, \dots, T$ is obtained from

$$\dot{Y}_{it} \text{ on } 1, D_i, i = 1, \dots, N \quad (2.19)$$

In order to conduct small N (or small N_1 inference), we cannot rely on the central limit theorem in the post-treatment period. Also, we cannot easily obtain standard errors for linear combinations for the $\hat{\tau}_{t,DD}$ because their sampling errors may be correlated in complicated ways due to serial dependence. (Unless we make the ideal assumptions that rule out serial correlation.) Nevertheless, we can obtain a CI for each $\tau_t = E[Y_{it}(1) - Y_{it}(0) | D_i = 1]$.

The coefficients one obtains from (2.19) are identical to running a standard panel DiD using the control and treatments for the time periods $\{1, 2, \dots, S-1, t\}$ where $S \leq t \leq T$. That is, we include all pre-treatment data.

Again, with N large enough, we can add controls $\mathbf{X}_i$. Note that, like the treatment effects themselves, the intercept and slope parameters on $\mathbf{X}_i$ vary freely over the treatment periods.

2.6 Using Different Pre-Treatment Periods

The suggestion in the previous subsection is to use all pre-treatment periods in computing the pre-treatment averages. Under NA and parallel trends, that approach uses the most information. Nevertheless, one might want to ignore information in the pre-treatment time periods. The standard event-study approach uses only the period just before the treatment. This results in the transformation

$$\dot{Y}_{it} = Y_{it} - Y_{i,S-1}, \quad t = S, \dots, T, \quad (2.20)$$

which can then be used in place of $\dot{Y}_{it}$ in (2.19). This is the transformation used by [Callaway and Sant'Anna \(2021\)](#). Of course, one can use any average of $\{Y_{it} : t = 1, \dots, S-1\}$ – even a weighted average – and the method would be still valid. If there is concern about violation of no anticipation, the average $\bar{Y}_{i,pre}$ could be replaced with

$$\bar{Y}_{i,S_0} = \frac{1}{S_0} \sum_{r=1}^{S_0} Y_{ir} \quad (2.21)$$

where $S_0 < S-1$. A long-difference also can be used: $Y_{it} - Y_{i,S_0}$.

3 Extensions: Heterogeneous Trends, Seasonality, and Aggregated Outcomes

The central contribution of the previous section concerns inference after a panel has been transformed into a cross section. This section considers alternative transformations motivated by different models for untreated outcomes. In particular, unit-specific detrending is an identification device rather than a separate inferential principle. Once a transformation has been chosen, inference again proceeds from a cross-sectional regression and requires the corresponding assumptions on its transformed errors.

3.1 Unit-Specific Linear Trends

LW (2026) show that unit-specific pre-treatment trends can be removed before estimating treatment effects. For concreteness, we first consider linear trends.

Procedure 3.1 (Unit-Specific Detrending):

1. For each unit i , estimate its pre-treatment linear trend from

$$Y_{it} \text{ on } 1, t, \quad t = 1, \dots, S-1, \quad (3.1)$$

and denote the estimated coefficients by $\hat{A}_i$ and $\hat{B}_i$.

2. For the post-treatment periods, form the out-of-sample residuals

$$\ddot{Y}_{it} = Y_{it} - \hat{Y}_{it} \equiv Y_{it} - \hat{A}_i - \hat{B}_i t, \quad t = S, \dots, T, \quad (3.2)$$

where $\hat{Y}_{it} = \hat{A}_i + \hat{B}_i t$ is the extrapolated pre-treatment trend.

3. Average the adjusted post-treatment outcomes:

$$\bar{\ddot{Y}}_i \equiv \frac{1}{T-S+1} \sum_{t=S}^T \ddot{Y}_{it}. \quad (3.3)$$

4. Estimate the average effect, $\hat{\tau}_{DT}$, from

$$\bar{\ddot{Y}}_i \text{ on } 1, D_i, \quad i = 1, \dots, N. \quad (3.4)$$

The identifying content of Procedure 3.1 is different from standard parallel trends. The procedure is useful when systematic differences in the untreated outcome paths can be adequately represented by unit-specific linear trends estimated before treatment. After those trends are removed, the remaining transformed untreated outcomes must be comparable across the treated and control groups. Thus, detrending should not be interpreted as a general solution to violations of parallel trends; it imposes a particular parametric model for those violations.

The inference step is the same one developed in Section 2. For example, exact inference with $N_1 = 1$ is available if

$$\bar{\ddot{Y}}_i = \alpha + \tau_{DT} D_i + U_i, \quad U_i \mid D_i \sim \text{Normal}(0, \sigma^2),$$

with independence across i . HC3 or RI can be considered under the alternative conditions described in Table 1. Consequently, any improvement in point estimation from detrending should be distinguished from the performance of the inference procedure applied after detrending. We make this distinction explicit in the Monte Carlo analysis.

There is no incidental-parameters problem from estimating $(\hat{A}_i, \hat{B}_i)$ separately for each unit. The final regressand $\bar{\ddot{Y}}_i$ is simply a linear function of that unit's time series. If the original units are independent across i , the transformed cross-sectional observations remain independent across i . The exact distributional conditions, however, are conditions on the resulting transformed errors and do not follow automatically from this independence.

Separate effects can also be estimated for each post-treatment period. For each $t = S, \dots, T$, regress

$$\ddot{Y}_{it} \text{ on } 1, D_i, \quad i = 1, \dots, N.$$

The same identifying and inferential qualifications apply.

3.2 Alternative Trend and Seasonal Adjustments

Linear detrending is only one possible transformation. When there are enough pre-treatment periods, the pre-treatment projection can include higher-order terms such as t^2 or t^3 . More flexible detrending can reduce systematic pre-treatment differences, but it also uses additional degrees of freedom and may remove substantial outcome variation.

One can alternatively use a second-difference-type transformation. For example,

$$\ddot{\ddot{Y}}_{it} \equiv (Y_{it} - Y_{i,S-1}) - (Y_{i,S-1} - Y_{iR}), \quad 1 \leq R < S - 1.$$

This is a difference-in-difference-in-differences transformation. Because it uses only a small number of periods rather than averages over longer pre- and post-treatment windows, the transformed outcome may be more sensitive to deviations from normality.

For quarterly, monthly, or other higher-frequency data, unit-specific seasonal indicators can be included in the pre-treatment projection, possibly together with a trend. The resulting deseasonalized or detrended outcomes can then be passed to the same cross-sectional estimation and inference step. Again, the transformation itself does not guarantee the conditions needed for exact inference.

3.3 Aggregated and Repeated Cross-Sectional Outcomes

The same transformations can be applied when the panel outcome is itself an aggregate of lower-level observations, but additional care is required. For example, suppose treatment is assigned at the state level and the underlying data are repeated cross sections of individuals.

A state-year outcome may be constructed as

$$\bar{Y}_{st} = \sum_{i \in (s,t)} w_{ist} Y_{ist}, \quad \sum_{i \in (s,t)} w_{ist} = 1.$$

Equal weights, survey weights, or other design weights may be used as appropriate.

Stacking $\{\bar{Y}_{st} : t = 1, \dots, T\}$ produces a state-level panel to which the transformations in Sections 2 and 3 can be applied. However, aggregation does not by itself justify the exact-inference assumptions. Sampling variation in $\bar{Y}_{st}$ may differ across states or years, particularly when cell sizes vary, and this can induce heteroskedasticity in the transformed state-level errors. For the usual OLS t statistic to have an exact finite-sample t distribution, the transformed state-level errors must therefore satisfy the same normality, homoskedasticity, and cross-unit independence conditions imposed above.

When the lower-level observations are randomly sampled within each state-period cell and cell sizes are sufficiently large, the sampling error in the estimated cell means may be small and approximate normality may be plausible. In other settings, the sampling variation should not be treated as negligible. Thus, the applicability of the procedure to repeated cross-sectional data depends not only on constructing the aggregate panel but also on whether the resulting transformed group-level outcomes support the chosen inference method.

For quarterly data, for example, one may construct

$$\bar{Y}_{stq} = \sum_{i \in (s,t,q)} w_{istq} Y_{istq}, \quad \sum_{i \in (s,t,q)} w_{istq} = 1, \quad q \in \{1, 2, 3, 4\},$$

and then include quarter indicators in the unit-specific transformation. The same inferential qualifications apply.

4 Comparison with Synthetic Control Methods

Synthetic control (SC) and, more recently, synthetic difference-in-differences (SDiD) have become widely used methods in the small- N settings that motivate this paper, particularly

when there is one or only a small number of treated units. Because these settings are also the primary focus of our approach, it is useful to compare our transformation-based procedures with SC and SDiD and to clarify the circumstances under which each method may be particularly attractive. We view the approaches as complementary rather than competing, because they use different sources of structure to construct the counterfactual and can perform well under different data configurations.

With a single treated unit, ADH (2010) propose algorithms that use pre-treatment data to obtain weights on “donors” to construct a synthetic control (SC) for the treated unit. The donor weights are chosen by an optimization algorithm, subject to being nonnegative and summing to one, so that the synthetic control reproduces the treated unit’s pre-treatment outcomes and other predictors as closely as possible. In early SC applications, inference is largely placebo-based: each control unit is treated in turn as if it had received the intervention, and the post-treatment gap for the actual treated unit is compared with the corresponding placebo gaps.

More recently, Arkhangelsky, Athey, Hirshberg, Imbens and Wager (2021) propose synthetic difference-in-differences (SDiD), which combines features of synthetic control and difference-in-differences. Unlike standard synthetic control, SDiD does not require the treated and synthetic-control outcome paths to match exactly in levels, allowing for a persistent level difference between them. The estimator can be interpreted as a weighted two-way fixed effects (TWFE) estimator: it assigns unit weights to balance the pre-treatment outcome paths of treated and control units and time weights to balance pre- and post-treatment periods within the control group. The unit and time weights are obtained from regularized optimization problems.

The key distinction for our purposes is that SC and SDiD use pre-treatment outcomes to construct a data-driven counterfactual rather than imposing the conventional parallel-trends assumption. This flexibility can be particularly valuable when untreated outcome paths are irregular or difficult to summarize with a simple parametric trend, provided that the donor pool is sufficiently rich to reproduce those paths.

Our approach takes a different route. Rather than estimating donor weights, we first transform the outcomes to remove the specified structure in untreated outcome evolution and then form a simple comparison path. Under Procedure 2.1, the cross-sectional average

of the transformed controls,

$$\left\{ N_0^{-1} \sum_{i=1}^{N_0} \dot{Y}_{it} : t = 1, \dots, T \right\}, \quad (4.1)$$

serves as the comparison path for the average transformed outcomes of the treated units,

$$\left\{ N_1^{-1} \sum_{i=N_0+1}^{N_0+N_1} \dot{Y}_{it} : t = 1, \dots, T \right\}. \quad (4.2)$$

If unit-specific trends are removed, $\ddot{Y}_{it}$ replaces $\dot{Y}_{it}$; systematic seasonal components can be handled similarly. Unlike SC and SDiD, no additional donor weights are estimated after the transformation, so all included controls enter the comparison with equal weight.

This feature is especially useful when the donor pool is small. Although the transformation-based approach is not limited to such settings, it may be particularly attractive when only a few credible controls are available. In that case, synthetic weighting may place most of the weight on only a few donors, whereas our approach uses all available controls after transformation. This situation arises in practice. In [Hur, Lee and Wooldridge \(2026\)](#), we revisit an existing empirical application with only three available control units, while the application in this paper considers a setting with only four. Together, the two applications illustrate the usefulness of the transformation-based approach when credible comparison units are available only in very small numbers.

The inferential logic of SDiD differs from the finite-sample argument in [Section 2](#). The theoretical guarantees for SDiD are developed using asymptotic approximations involving the number of control units and the lengths of the pre- and post-treatment periods. Our exact t procedure instead conditions on a finite cross-sectional sample and obtains an exact t distribution by imposing normality and homoskedasticity directly on the transformed errors. These are different inferential tradeoffs. The cross-sectional approach does not require large N_0 , N_1 , T_0 , or T_1 for the finite-sample calculation itself, but the exact result depends on stronger distributional assumptions.

We therefore view the proposed methods as complementary to SC and SDiD rather than as substitutes. Which approach is most attractive depends on the size and informativeness of the donor pool, the amount of pre-treatment information, the structure of untreated outcome heterogeneity, and the credibility of the assumptions used for estimation and inference.

5 Monte Carlo Simulations

We use Monte Carlo simulations to study the finite-sample performance of alternative inference procedures when the number of treated and/or control units is small. The purpose of the exercise is to isolate inference from point estimation. In the common-timing design considered here, the proposed demeaning estimator is numerically equivalent to the usual DiD estimator. Thus, the methods compared below use the same underlying treatment-effect estimate and differ only in how uncertainty is quantified.

5.1 Small- N Inference

We consider a balanced panel with $N = N_0 + N_1$ independent units observed over $T = T_0 + T_1$ periods. Treatment begins in period $T_0 + 1$ for all treated units and remains in place thereafter. In each replication, exactly N_1 units are randomly assigned to treatment, independently of the potential outcomes.

Potential outcomes in control state are generated as

$$Y_{it}(0) = \alpha_i + \lambda_t + U_{it}, \quad (5.1)$$

where

$$\alpha_i \sim N(0, 1) \quad \text{and} \quad \lambda_t = 0.05t + 0.30 \sin(t/2).$$

The common time effect is deliberately nonlinear, but it is shared by treated and control units. Because treatment is randomly assigned independently of the potential outcomes, parallel trends holds by construction.

In the baseline design, the idiosyncratic component follows a stationary Gaussian AR(1) process,

$$U_{i1} \sim N\left(0, \frac{1}{1 - \rho^2}\right), \quad (5.2)$$

$$U_{it} = \rho U_{i,t-1} + \varepsilon_{it}, \quad \varepsilon_{it} \sim N(0, 1), \quad t = 2, \dots, T. \quad (5.3)$$

We set $\rho = 0.75$ in the baseline design. Thus, outcomes may exhibit substantial serial correlation within unit, while units remain independent across the cross section.

Observed outcomes are

$$Y_{it} = Y_{it}(0) + \tau W_{it}. \quad (5.4)$$

We apply Procedure 2.1 to each simulated panel. Specifically, we use the pre-treatment observations to demean each unit’s post-treatment outcomes and then average the resulting out-of-sample residuals over the post-treatment periods. This yields one transformed observation per unit,

$$\bar{Y}_i = \Delta \bar{Y}_i,$$

and we estimate the cross-sectional regression

$$\Delta \bar{Y}_i = \alpha + \tau D_i + V_i. \quad (5.5)$$

Under the baseline DGP, V_i is independent, normal, and homoskedastic across units. The usual t statistic from (5.5) therefore has its exact finite-sample distribution even though the original panel outcomes are serially correlated within unit.

We consider two inference procedures based on this transformed cross-sectional regression. *Exact t* uses the conventional OLS standard error and the $\mathcal{T}_{N-2}$ reference distribution. *HC3* instead uses the HC3 heteroskedasticity-robust variance estimator. Because HC3 is not available when one treatment arm contains only a single observation, it is omitted for designs with $N_1 = 1$ or $N_0 = 1$. We also report randomization inference (RI), which reassigns treatment across units while holding the number treated fixed.

Panel-regression benchmarks. For comparison, we also estimate the conventional two-way fixed effect (TWFE) panel regression

$$Y_{it} = \alpha_i + \lambda_t + \tau W_{it} + e_{it}, \quad (5.6)$$

and cluster standard errors by unit. We refer to this procedure as *TWFE cluster*. As an additional benchmark for size in selected designs, we also report wild cluster bootstrap (WCB) inference based on the same TWFE specification.

Because the point estimator is common across the procedures, our main outcomes are inferential. When $\tau = 0$, we report the empirical rejection probability of a nominal five-percent test. When $\tau = 1$, we report the empirical coverage probability of a nominal 95-percent confidence interval for procedures that directly produce confidence intervals. For RI and WCB, we report rejection probabilities under the null.

The main simulations vary the sizes of the treated and control groups separately. We first fix $N_0 = 20$ and vary the number of treated units. We then fix $N_1 = 5$ and vary

the number of control units. We supplement these main exercises with sensitivity analyses that vary within-unit serial correlation, panel length, the innovation distribution, and cross-sectional heteroskedasticity. Table 2 summarizes the simulation designs.

Table 2: Monte Carlo designs for small- N inference

Exercise	Parameter varied	Design
Few treated units	N_1	$N_0 = 20, N_1 \in \{1, 2, 5, 10\}$
Few control units	N_0	$N_1 = 5, N_0 \in \{2, 5, 10, 20\}$
Serial correlation	ρ	0.75 (baseline); 0, 0.5, 0.9 (sensitivity)
Panel length	$T_0 = T_1$	$T_0 = T_1 \in \{3, 10, 20\}$
Error distribution	ε_{it}	[1] Normal [2] Student- t_5 : $\varepsilon_{it} = T_{5,it}/\sqrt{5/3}$ [3] χ^2_2 : $\varepsilon_{it} = (\chi^2_{2,it} - 2)/2$
Heteroskedasticity	σ_i	Baseline: $\sigma_i = 1$ Heteroskedastic: $\sigma_i = \exp(0.75Z_i - 0.75^2/2)$, $Z_i \sim N(0, 1)$

Note: Unless otherwise stated, the baseline design sets $T_0 = T_1 = 10$ and $\rho = 0.75$.

Note: The serial-correlation and panel-length exercises use $N_0 = 20$ and $N_1 = 2$. The non-normal designs are evaluated at $T_0 = T_1 \in \{3, 20\}$ with $N_0 = 20$ and $N_1 = 2$, while the heteroskedasticity exercise uses $N_0 = 20$ and $N_1 = 5$.

The serial-correlation and panel-length exercises examine *sensitivity* within the baseline Gaussian homoskedastic setting. The non-normal and heteroskedastic designs instead serve as stress tests of the distributional conditions underlying exact inference.

5.2 Simulation Results

Table 3 reports the main results for the *few treated units* design summarized in the first row of Table 2. Because the point estimate is common across procedures, we focus on empirical size, measured by the rejection rate when the true treatment effect is zero, and on 95-percent confidence-interval coverage when the true effect equals one. For RI and WCB, which are implemented here only for testing $H_0 : \tau = 0$, we report empirical rejection probabilities.

Table 3: Inference with Few Treated Units

N_1	Exact t		HC3		TWFE cluster		RI	WCB
	Size	Coverage	Size	Coverage	Size	Coverage	Size	Size
1	0.050	0.945	–	–	0.658	0.360	0.030	0.000
2	0.043	0.958	0.142	0.862	0.243	0.774	0.041	0.000
5	0.050	0.941	0.061	0.936	0.085	0.915	0.052	0.061
10	0.048	0.953	0.043	0.958	0.050	0.954	0.047	0.047

Note: $N_0 = 20$, $T_0 = T_1 = 10$, $\rho = 0.75$, and errors are normal. HC3 is unavailable when $N_1 = 1$.

Note: Results are based on 2,000 Monte Carlo replications, with 999 randomization or bootstrap replications for RI and WCB, respectively.

The results highlight the importance of the size of the treated group for inference. The exact t procedure remains close to the nominal rejection rate and confidence-interval coverage even when very few units are treated. By contrast, conventional unit-clustered TWFE inference can be substantially distorted when the treated group is very small. When $N_1 = 1$, for example, the rejection rate for TWFE cluster is 0.658, compared with 0.050 for the exact t procedure, while its 95-percent coverage falls to 0.360. With $N_1 = 2$, TWFE cluster continues to over-reject and under-cover, although the distortions are less severe.

As the number of treated units increases, the differences across procedures narrow considerably. By $N_1 = 10$, both the exact t procedure and unit-clustered TWFE inference are close to their nominal levels. Thus, having many control units does not by itself ensure reliable clustered inference when the number of treated units is small.

The additional sensitivity analyses lead to similar conclusions. The exact t procedure remains close to nominal size and coverage across different degrees of serial correlation and panel lengths. It also performs reasonably well under the non-normal error distributions considered here, although exact finite-sample validity is no longer guaranteed when normality is relaxed. Complete results for the additional designs are reported in Appendix [B](#).

6 Revisiting the California Cigarette Tax Application

We revisit the canonical California tobacco-control application of ADH ([2010](#)) using our proposed methods. The first year of the program is 1989, with 19 pre-treatment years

(1970-1988) and 12 treatment years (1989-2000). We use the log of per capita cigarette sales as the outcome variable; ADH (2010) used the level of this variable, but the log seems more natural – partly to obtain a percentage effect and partly to make normality of the transformed outcome a better approximation. California is the only treated state. As in ADH (2010), we initially use $N_0 = 38$ potential control states after eliminating states that implemented some sort of anti-smoking program over the period.

As discussed in the previous section, SCM uses pre-treatment outcomes (and sometimes other variables) to create a synthetic control – in this case, a synthetic California. The weights for the 38 controls are chosen to make the SC as close to California as possible based on pre-treatment observables (primarily Y_{it}). These weights are then used to project out what would have been the untreated outcome for CA after the policy change; this is then compared with the actual outcome in post-intervention.

6.1 Results

Figure 1 plots the comparison paths defined in (4.1) and (4.2) using Procedure 2.1. The solid pink line represents the residualized outcome for the single treated unit, California, while the dashed blue line represents the average residualized outcome across the control states. The vertical dashed line marks the intervention year, 1989.

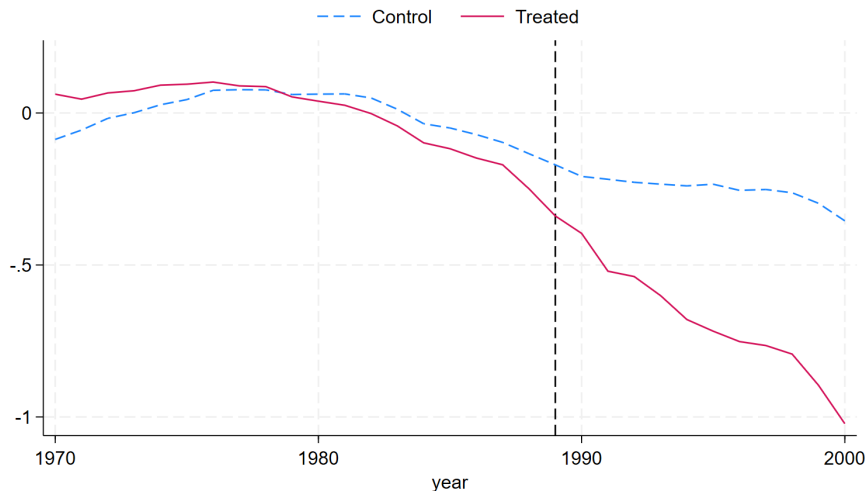


Figure 1: Removing Pre-Treatment Averages, All States as Control

The average of the controls does not track California particularly well before the intervention, indicating that removing only the pre-treatment mean is not sufficient.

By contrast, Figure 2 shows the residualized paths after removing state-specific linear trends using Procedure 3.1. The treated and control paths track each other much more closely before the policy change.

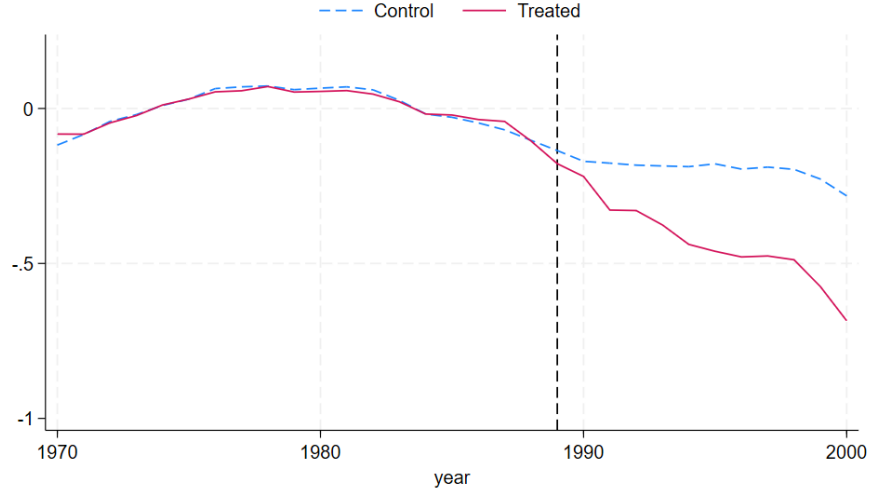


Figure 2: Removing Pre-Treatment Linear Trends, All States as Control

Figure 3 plots the gap between detrended California and the average of the 38 detrended control states shown in Figure 2. The gap remains close to zero throughout the pre-treatment period. After the 1989 intervention, California declines more sharply than the controls, consistent with a negative policy effect.

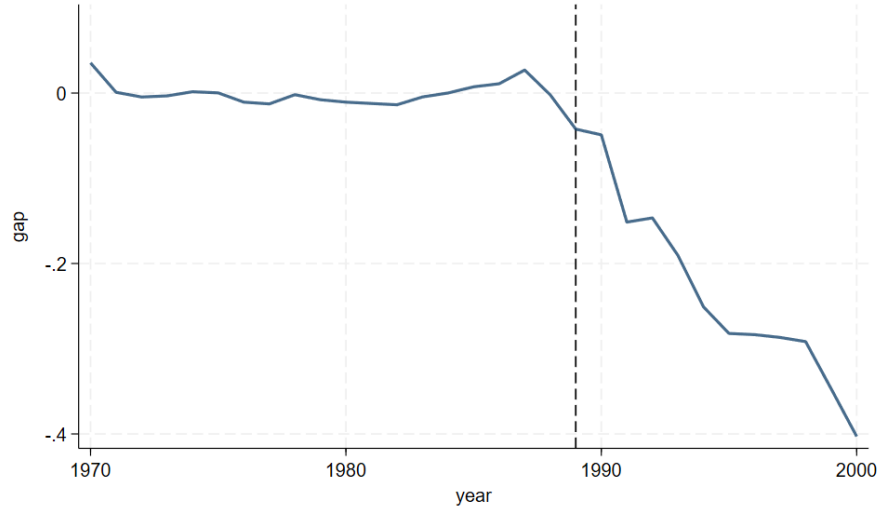


Figure 3: Gap Between California and Average of All Controls, State-Specific Linear Trend

Similarly, both the synthetic control and synthetic DiD methods, using all 38 states

as donor pool, display a similar pre-treatment trend between California and synthetic California (Control), as visualized in Figure 4 and Figure 5, respectively.

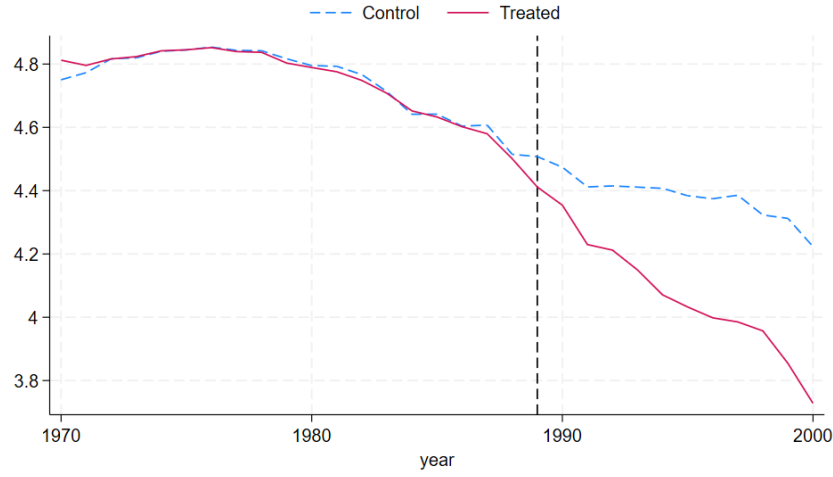


Figure 4: Synthetic Control Method, All States as Donor Pool

In Figure 4, the synthetic control method assigns positive weights to six of the 38 donor states—Nevada (0.31), Montana (0.30), Utah (0.28), Connecticut (0.06), Colorado (0.03), and New Hampshire (0.02)—to construct the synthetic California, with values in parentheses indicating the assigned weights. In contrast, our detrending procedure incorporates information from all 38 states in the donor pool.

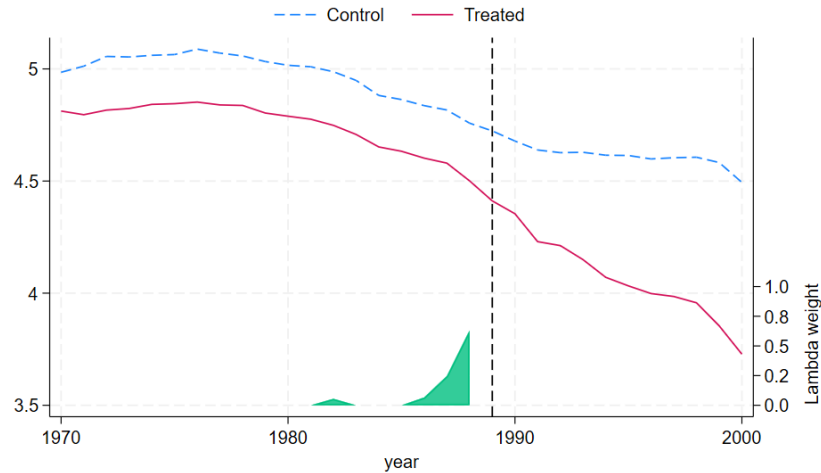


Figure 5: Synthetic DiD, All States as Donor Pool

Figure 5 presents the synthetic DiD results. Similar to the synthetic control method,

only a small subset of donor states receives positive unit weights. Unlike synthetic control method, however, SDiD allows for a persistent level difference between the treated and synthetic-control outcome paths. To account for this difference in the DiD comparison, SDiD also estimates time weights that determine the weighted pre-treatment gap used as the benchmark for the post-treatment comparison. These weights, shown in green, are concentrated in the mid- to late-1980s.

Table 4 reports estimated effects and standard errors from four methods: (i) demeaning (Procedure 2.1), (ii) detrending (Procedure 3.1), (iii) synthetic control, and (iv) synthetic DiD. The first row reports the average effect over all treated periods, with subsequent rows showing period-specific effects. The estimated effect becomes more negative over time, reaching -0.403 by 2000.

Table 4: Estimated ATTs using 38 states as the donor pool

	Procedure 2.1 (DiD)	Procedure 3.1 (Unit-Specific Detrending)	SC	Synthetic DiD
Average Effect	-0.422*** (0.121)	-0.227** (0.094)	-0.304*** (0.112)	-0.286*** (0.097)
τ_{1989}	-0.168* (0.096)	-0.043 (0.059)		
τ_{1995}	-0.484*** (0.137)	-0.282** (0.112)		
τ_{2000}	-0.667*** (0.164)	-0.403** (0.152)		

Note 1: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

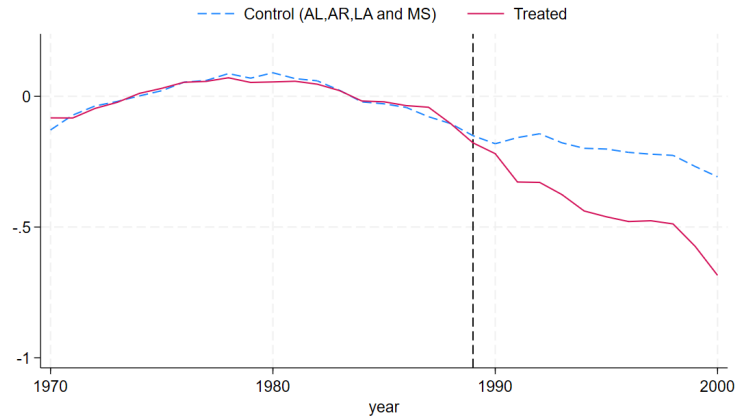
Note 2: For Procedure 3.1, the exact-inference p -value (under normality) is 0.021, and the randomization-inference p -value (1,000 replications) is 0.020. The two are nearly identical.

6.2 Four Southern States as the Donor Pool

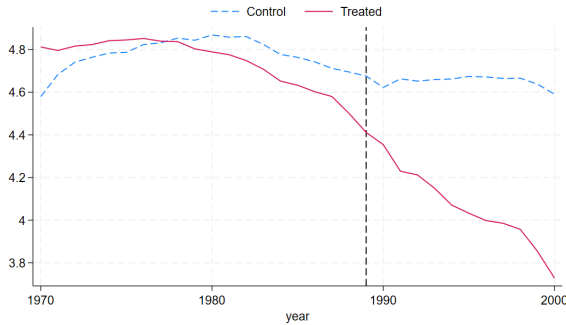
To further assess the robustness of our state-specific detrending procedure in this application, we compare the results from Procedure 3.1 with those from synthetic control and synthetic DiD methods, restricting the donor pool to a small set of states that, a priori, do not appear “similar” to California. While the previous section used all 38 states as donor pool, here we focus on a smaller donor pool consisting of four Southern states —Alabama (AL), Arkansas (AR), Louisiana (LA), and Mississippi (MS)— which differ substantially from California in socioeconomic characteristics.

By limiting the set of donor states in this way, we evaluate how each estimator performs when the availability of donor units is limited, thereby providing additional insights into the robustness of our procedure. For completeness, results based on an alternative donor pool of Midwestern states —Illinois (IL), Iowa (IA), Minnesota (MN) and Ohio (OH)— are presented in Appendix A.

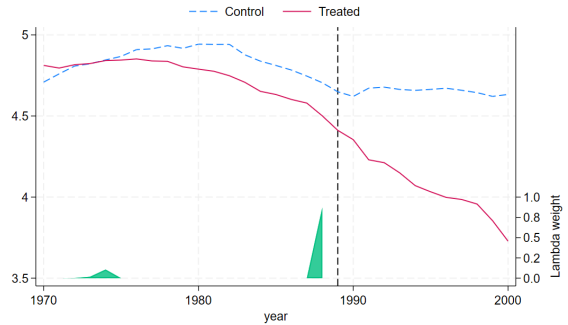
Figure 6 illustrates the results using four Southern states as the donor pool. Panel (a) shows that our Procedure 3.1 (unit-specific detrending) achieves a similarly strong pre-treatment fit even with only four control states, closely matching the fit obtained using the full set of 38 controls.



(a) State-Specific Detrending



(b) Synthetic Control



(c) Synthetic DiD

Figure 6: AL, AR, LA, and MS as Donor Pool

By contrast, as shown in Panel (b) and (c) of Figure 6, both the synthetic control and synthetic DiD methods fail to recover a convincing synthetic California. They exhibit

distinct pre-period patterns, with California’s cigarette sales declining faster than those of the synthetic California (Control) after the mid-1970s.

Table 5 summarizes the estimated ATTs using AL, AR, LA, and MS as the donor pool. The estimates from Procedure 3.1 closely mirror those obtained when all 38 states are used as the donor pool, both in magnitude and statistical significance. In contrast, the synthetic control and synthetic DiD methods yield estimates of larger absolute value, reflecting the poorer pre-treatment fits observed in Figure 6. This pattern likely arises because California’s cigarette sales were already declining more rapidly prior to the intervention, contaminating the post-treatment estimates of these methods.

Table 5: Estimated ATTs using AL, AR, LA, MS as the donor pool

	Procedure 2.1 (DiD)	Procedure 3.1 (Unit-Specific Detrending)	SC	Synthetic DiD
Average	-0.556*** (0.080)	-0.215** (0.039)	-0.571*** (0.034)	-0.392*** (0.030)
1989	-0.247 (0.107)	-0.027 (0.052)		
1995	-0.611*** (0.077)	-0.259** (0.055)		
2000	-0.839*** (0.032)	-0.377** (0.115)		

Note: standard errors in parentheses: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

7 Staggered Rollouts

We can extend the previous methods to the case of staggered interventions, where units are allowed to be first treated at different times. But one must use care in choosing a suitable control group.

7.1 The Setup and Estimation

Suppose now that the first treatment period is S , but treatment can first occur in periods $S, S + 1, \dots, T$. Index the treatment cohorts or groups by $g \in \{S, \dots, T\}$, the time of first treatment. Let D_g , $g = S, S + 1, \dots, T$ be the cohort indicators. Specifically, $D_{ig} = 1$ if and only if unit i was first treated in period g . The never treated units are indicated by $D_{i\infty} = 1$.

There are N_g units in each cohort, with N_∞ being the number of never treated units.

With $N_\infty \geq 2$, the other treated cohorts may have only one unit. Naturally, if there are periods without new treated units, there will be no treatment effects estimated in those periods.

The potential outcomes are $Y_t(g)$, $g = \{S, \dots, T, \infty\}$, where $Y_t(g)$ is the outcome in period t if the first period of treatment is g . $Y_t(\infty)$ is the never treated state (or control state). The treatment effects are now indexed by cohort and time:

$$\tau_{gt} = E[Y_t(g) - Y_t(\infty) | D_g = 1], \quad t = g, \dots, T. \quad (7.1)$$

We are also interested in estimating the cohort-specific average effects,

$$\tau_g = \frac{1}{(T - g + 1)} \sum_{t=g}^T \tau_{gt}. \quad (7.2)$$

As is common, we make a no anticipation Assumption:

$$Y_t(g) = Y_t(\infty), \quad t < g,$$

which says that all potential outcomes are the same as in the never-treated state in periods before the treatment occurs.

For a given cohort g , one now removes the average or a trend using data up through $g - 1$, the final pre-treatment period. It is easiest to obtain the transformation for each unit in the sample, and then sort out the valid control units afterwards. For each i , one runs the regressions

$$Y_{it} \text{ on } 1, \quad t = 1, 2, \dots, g - 1 \quad (7.3)$$

Because we have different treatment cohorts, we add a cohort subscript when projecting out to post-treatment periods:

$$\dot{Y}_{itg} = Y_{it} - \bar{Y}_{i,pre(g)}, \quad t = g, \dots, T \quad (7.4)$$

where $\bar{Y}_{i,pre(g)}$ is the average of Y_{it} for $t = 1, \dots, g - 1$. To remove a linear time trend, the regressions are

$$Y_{it} \text{ on } 1, t, \quad t = 1, 2, \dots, g - 1 \quad (7.5)$$

and then the out-of-sample residuals are

$$\ddot{Y}_{itg} = Y_{it} - \hat{A}_{ig} - \hat{B}_{ig} \cdot t, \quad t = g, \dots, T \quad (7.6)$$

Once the transformed variables have been obtained, the key is choosing a valid control group at time t for treatment cohort g (which could have as few as one unit). One can use only the never treated units at time t , provided $N_\infty \geq 2$. However, as shown in LW (2026), under suitable NA and PT assumptions, one can use any unit not-yet treated (NYT). Then for cohort g in time period t , the units represented by

$$D_{i,t+1} + D_{i,t+2} + \cdots + D_{iT} + D_{i\infty} = 1 \quad (7.7)$$

are valid controls. For efficiency reasons, one should use all possible controls. Nevertheless, one can use any subset (including the NT group). Let $C_{i,t+1}$ be the control group chosen for period t , with the subscript indicating that these groups cannot have been treated prior to $t + 1$. Then $\hat{\tau}_{gt}$ is the coefficient on D_{ig} in the cross-sectional regression

$$\dot{Y}_{itg} \text{ on } 1, D_{ig} \text{ using } D_{ig} + C_{i,t+1} = 1 \quad (7.8)$$

Or, replace $\dot{Y}_{itg}$ with the detrended version, $\ddot{Y}_{itg}$. Under a homoskedastic normality assumption, exact inference can be used in (7.8) even if $N_g = 1$ and the number of control units is small.

For many purposes, we are more interested in the τ_g , which we can aggregate to obtain a single, weighted effect – more on this below. Then, because we want inference without many units in the treated cohort, we use the never treated units as the control and run a single, aggregated regression. Specifically, define

$$\bar{\dot{Y}}_{ig} \equiv \frac{1}{(T - g + 1)} \sum_{t=g}^T \dot{Y}_{itg} \quad \text{and} \quad \bar{\ddot{Y}}_{ig} \equiv \frac{1}{(T - g + 1)} \sum_{t=g}^T \ddot{Y}_{itg}. \quad (7.9)$$

Then run the regression

$$\bar{\dot{Y}}_{ig} \text{ on } 1, D_{ig}, i = 1, \dots, N, D_{ig} + D_{i\infty} = 1 \quad (7.10)$$

to obtain $\hat{\tau}_g$ as the coefficient on D_{ig} , or use $\bar{\ddot{Y}}_{ig}$ in place of $\bar{\dot{Y}}_{ig}$. Again, this is a cross-sectional regression with independent observations. If N_g and N_∞ are suitably large to employ asymptotics, then we can use a heteroskedasticity-robust standard error (HC3) to obtain a t statistic or confidence intervals. We can appeal to exact theory if N_g is small, including $N_g = 1$.

If we want to aggregate the $\hat{\tau}_g$ to obtain a single treatment effect, a challenge is to obtain a valid standard error when cohort sizes are not particularly large. A natural parameter is to weight the cohort ATTs, τ_g , but the cohort shares. This leads to the estimator

$$\hat{\tau}_\omega = \sum_{g=S}^T \hat{\omega}_g \hat{\tau}_g \quad (7.11)$$

where

$$\hat{\omega}_g = \frac{N_g}{N_S + N_{S+1} + \dots + N_T}, \quad (7.12)$$

which represent the cohort shares out of the treated cohorts. A standard error for $\hat{\tau}_\omega$ must account for the covariances among the $\hat{\tau}_g$, which is difficult if we cannot appeal to asymptotics. Even with a somewhat larger number of total treated units, a trick is helpful. We know from simple regression analysis that $\hat{\tau}_g$ is a difference in means:

$$\hat{\tau}_g = \frac{1}{N_g} \sum_{i=1}^N D_{ig} \bar{Y}_{ig} - \frac{1}{N_\infty} \sum_{i=1}^N D_{i\infty} \bar{Y}_{ig} \quad (7.13)$$

and so, by simple algebra,

$$\hat{\tau}_\omega = \frac{1}{(N_S + N_{S+1} + \dots + N_T)} \sum_{g=S}^T \sum_{i=1}^N D_{ig} \bar{Y}_{ig} - \frac{1}{N_\infty} \sum_{g=S}^T \sum_{i=1}^N D_{i\infty} \hat{\omega}_g \bar{Y}_{ig} \quad (7.14)$$

Define a treatment indicator

$$D_i = D_{iS} + \dots + D_{iT} \quad (7.15)$$

the number of treated units as

$$N_{treat} = N_S + N_{S+1} + \dots + N_T.$$

Rearrange (7.14) to obtain

$$\begin{aligned} \hat{\tau}_\omega &= \frac{1}{N_{treat}} \sum_{i=1}^N D_i \cdot \left(\sum_{g=S}^T D_{ig} \bar{Y}_{ig} \right) - \frac{1}{N_\infty} \sum_{i=1}^N D_{i\infty} \cdot \left(\sum_{g=S}^T \hat{\omega}_g \bar{Y}_{ig} \right) \\ &\equiv \frac{1}{N_{treat}} \sum_{i=1}^N D_i \cdot \bar{Y}_i - \frac{1}{N_\infty} \sum_{i=1}^N D_{i\infty} \cdot \bar{Y}_i \end{aligned} \quad (7.16)$$

$$= \frac{1}{N_{treat}} \sum_{i=1}^N D_i \cdot \bar{Y}_i - \frac{1}{N_{control}} \sum_{i=1}^N (1 - D_i) \cdot \bar{Y}_i \quad (7.17)$$

where

$$\bar{Y}_i \equiv D_{iS} \cdot \bar{Y}_{iS} + \dots + D_{iT} \cdot \bar{Y}_{iT} + D_{i\infty} \cdot \left(\sum_{g=S}^T \hat{\omega}_g \bar{Y}_{ig} \right). \quad (7.18)$$

Note that the representation in (7.17) uses the fact that $D_i \cdot D_{ig} = D_{ig}$. Also, $D_i + D_{i\infty} = 1$, and so (7.17) is a simple difference in sample mean of $\bar{Y}_i$ between units that are eventually treated and the never treated group. In other words, run the cross-sectional regression

$$\bar{Y}_i \text{ on } 1, D_i, \quad i = 1, \dots, N, \quad (7.19)$$

and then $\hat{\tau}_\omega$ is the coefficient on D_i . Naturally, if we use unit-specific detrending for each cohort date g to obtain the $\bar{\bar{Y}}_{ig}$, then we replace $\bar{Y}_{ig}$ with $\bar{\bar{Y}}_{ig}$ everywhere.

Obtaining $\hat{\tau}_\omega$ from the regression in (7.19) has some advantages. It automatically accounts for the correlations among the $\hat{\tau}_g$, and it can be used whether or not N_{treat} is small or large (and same for $N_{control}$). If N_{treat} and $N_{control}$ are even moderately large, a heteroskedasticity-robust standard error from (7.19) is justified via asymptotic analysis.

Arkhangelsky, Athey, Hirshberg, Imbens and Wager (2021) also discuss how SDiD can be applied to staggered interventions. They suggest splitting the sample by adoption date and applying the SDiD method separately to each treatment cohort, using the never treated cohort as the donor pool [see the appendix in AAHIW (2021)]. In other words, do exactly what we propose with our unit-specific demeaning or detrending. Therefore, our proposed methods and the way SDiD is implemented for staggered designs are directly comparable.

7.2 Application: Estimating the Effects of Expansions to Castle Doctrine on Homicides

We apply the proposed method to the data set distributed with Cunningham (2021) on changes in so-called “castle” laws —or “stand-your-ground” laws— and their impact on homicide rates in the United States. The treatment timing in the Cunningham data corresponds to the *statutory expansions* of self-defense protections studied by Cheng and Hoekstra (2013). In particular, these laws extended self-defense protections that traditionally applied within the home to locations outside the home, where individuals would no longer have a duty to retreat before using force in self-defense.

Importantly, the treatment indicator captures these statutory expansions during the sample period rather than the initial existence of castle doctrine or related self-defense protections in a state. States differed in their pre-existing self-defense laws and case-law protections before these statutory changes. Thus, throughout this application, “never treated” refers to states that are not coded as experiencing the relevant statutory expansion

during the 2000–2010 sample period, rather than states with no pre-existing castle-doctrine or stand-your-ground protections.

The data set covers all 50 U.S. states from 2000 through 2010. Under this coding, one state is treated in 2005, followed by 13 additional states in 2006, four states in 2007, two states in 2008, and one state in 2009. In total, there are $N_{\text{treat}} = 21$ eventually treated units and $N_{\text{control}} = 29$ states that are never treated during the sample period. The outcome variable is the logarithm of the annual number of homicides per 100,000 state residents.

Using the regression in (7.19) with $\bar{Y}_i$ defined in (7.18), the estimated aggregated treatment effect, $\hat{\tau}_\omega$, is about 0.092, or about 9.2% more homicides following the coded legal change. The usual OLS standard error is 0.057, which gives $t \approx 1.61$. This is not quite significant at the 10% level against a two-sided alternative. The HC3 t statistic is 1.50. Replacing $\bar{Y}_i$ with $\bar{\bar{Y}}_i$ — obtained from linear detrending — decreases the estimate to 0.067, the HC3 standard error is 0.055, and $t \approx 1.21$.

The SDiD estimate, again obtained using the `sdid` package in Stata 18, is 0.099 and the standard error, using the placebo method, is 0.069 ($t = 1.41$). Both the estimate and its precision are in close agreement with the demeaning method described in Section 7.1.

8 Additional Practical Considerations

8.1 Choosing the Number of Time Periods

With synthetic control-type approaches and the approaches we suggest here, one can study the robustness of the findings by adjusting the number of pre-treatment time periods. For example, when using the unit-specific detrending method, we can vary the starting point of the data as a way of evaluating how the estimates change as the unit-specific trends are removed using different stretches of data. This is in the spirit of [Rambachan and Roth \(2023\)](#), who formalize the idea of allowing a range of violations of parallel trends and studying the sensitivity to the estimates. In many examples, the policy intervention is likely based on past outcomes (in the untreated state). Does one need to go back, say, 20 years to remove unit-specific intercepts and trends to account for the selection into treatment? In many cases, fewer pre-treatment periods might suffice. Because our approach does not rely on large T_0 or T_1 (but does rely on normality), the robustness of the estimates can be studied by varying T_0 in particular.

8.2 Clustering and Spatial Correlation with Larger Cross Sections

Although the previous procedures are motivated by settings with few control or treated units, or few of both, our approach has benefits in settings where $N_{control}$ and N_{treat} are large enough to rely on large- N asymptotic analysis. Recall that the basic DiD estimator is obtained from

$$\bar{Y}_i \text{ on } 1, D_i, i = 1, \dots, N$$

and the one that removes pre-treatment unit-specific trends is

$$\ddot{Y}_i \text{ on } 1, D_i, i = 1, \dots, N.$$

We can even add controls with large enough N . In Section 7 we showed that the same regressions can be used in the case of staggered assignment by defining D_i to be an “ever treated” indicator – see (7.15) – and by modifying $\bar{Y}_i$ or $\ddot{Y}_i$ as in equation (7.18). Because these are cross-sectional regressions, it is straightforward to compute standard errors clustered at a level higher than i . For example, if i is a county, but we are studying a policy that varies only at the state level, we can cluster at the state level if we have a sufficiently large number of treated and control states. Again, it does not matter how large T_0 and T_1 are. Abadie, Athey, Imbens and Wooldridge (2023) discuss why standard errors should be clustered when the intervention is at a higher level than the unit. As another example, i could be a household whereas a policy is applied at the village level. Clustering can be done separately by time period, too. The key is that it is a cross-sectional regression and then the usual clustering methods can apply.

In situations with a spatial structure – for example, the assignment of treated units in implementing a new policy may be spatially correlated – we can obtain standard errors using a covariance matrix estimator robust to heteroskedasticity and spatial correlation. These so-called “SHAC” standard errors are proposed in Conley (1999).

9 Concluding Remarks

Building on Lee and Wooldridge (2026), we propose a simple transformation-based approach to inference in panel data settings with few treated or control units. In the common-timing case, the transformation reproduces the standard DiD estimator while recasting the

inference problem as a cross-sectional regression. This representation permits exact finite-sample inference under the classical linear model assumptions, with HC3 and randomization inference providing useful alternatives under different assumptions.

The same inferential framework can be applied after alternative transformations, including unit-specific detrending and seasonal adjustment, and extends naturally to staggered intervention timing and the estimation of period-specific effects. Thus, the choice of transformation governs the structure imposed on untreated outcome evolution, while inference proceeds from the resulting cross-sectional regression. The procedures can be readily implemented using our publicly available user-written Stata command, [lwdid](#).

Our Monte Carlo simulations show that the exact t procedure performs well when the number of treated or control units is small, while conventional unit-clustered TWFE inference can be substantially distorted with very few treated units. The procedure also performs well across a range of sensitivity designs, although exact finite-sample validity requires the maintained distributional assumptions.

In the California application, our unit-specific detrending produces estimates similar to synthetic DiD when the full donor pool is used, while continuing to perform well when the donor pool is restricted to only a few states, a setting in which SC and SDiD exhibit poor pre-treatment fit. In the staggered application to expansions of “castle” laws, the aggregated estimate based on unit-specific demeaning is also close to the corresponding synthetic DiD estimate. These applications highlight the practical advantages of the transformation-based approach, especially when only a few comparison units are available.

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Appendix

A Four Midwestern States as the Donor Pool

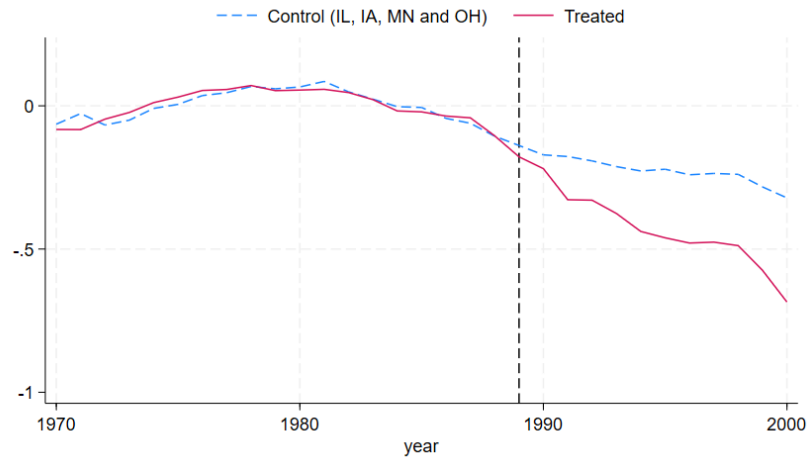
To complement the main robustness analysis using Southern states (Alabama, Arkansas, Louisiana, and Mississippi) as the donor pool, we replicate the exercise using an alternative donor pool consisting of four Midwestern states —Illinois (IL), Iowa (IA), Minnesota (MN), and Ohio (OH). The corresponding results are presented in Table 6.

Table 6: Estimated ATTs using IL, IA, MN and OH as the donor pool

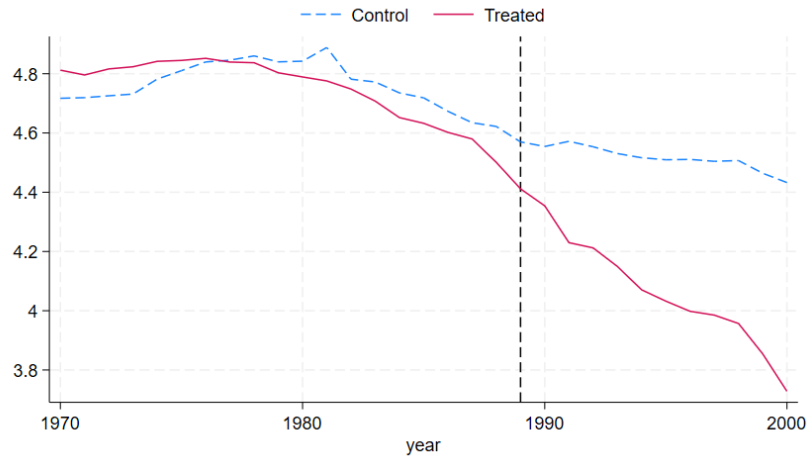
	Procedure 2.1 (DiD)	Procedure 3.1 (Unit-Specific Detrending)	SC	Synthetic DiD
Average	-0.413** (0.118)	-0.198* (0.079)	-0.437** (0.184)	-0.275* (0.154)
1989	-0.178* (0.071)	-0.040 (0.045)		
1995	-0.462** (0.133)	-0.239* (0.088)		
2000	-0.655** (0.183)	-0.363* (0.136)		
Note: standard errors in parentheses: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$				

Similar to the previous case with Southern states, panel (a) of Figure 7 shows that our detrending approach makes the control —defined as the average of residualized outcomes across all donor units— closely mimic the pre-treatment trajectory of California. In contrast, Panels (b) and (c) reveal the synthetic California (Control) constructed by the SC and SDiD methods trends differently from the actual California prior to the intervention, indicating poor pre-treatment fit.

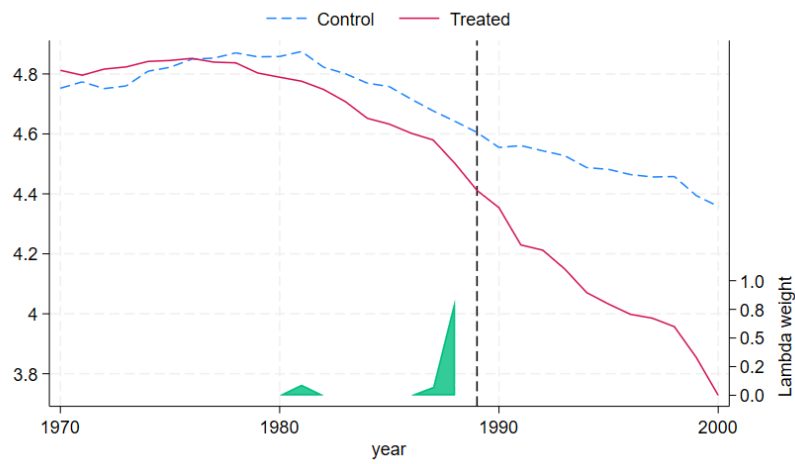
These results based on the Midwestern donor pool reinforce the robustness of our approach, even under limited-sample settings. In contrast to the synthetic control and synthetic DiD methods, which rely on weighted combinations of donor units, our procedure leverages all available donor states as controls through outcome transformation, ensuring stable estimates even with a small donor pool.



(a) State-Specific Detrending



(b) Synthetic Control



(c) Synthetic DiD

Figure 7: IL, IA, MN and OH as Donor Pool

B Additional Monte Carlo Results

This appendix reports the complete Monte Carlo results for the additional designs summarized in Table 2. All results are based on 2,000 Monte Carlo replications.

Table 7: Inference with Few Control Units

N_0	Exact t		HC3		TWFE cluster		RI	WCB
	Size	Coverage	Size	Coverage	Size	Coverage	Size	Size
2	0.049	0.949	0.048	0.954	0.088	0.906	0.028	0.040
5	0.048	0.942	0.031	0.961	0.048	0.943	0.042	0.055
10	0.041	0.951	0.037	0.960	0.050	0.942	–	–
20	0.053	0.947	0.061	0.938	0.087	0.913	–	–

Note: $N_1 = 5$, $T_0 = T_1 = 10$, $\rho = 0.75$, and errors are normal. RI and WCB are reported for $N_0 = 2$ and $N_0 = 5$ only. Results are based on 2,000 Monte Carlo replications, with 999 randomization or bootstrap replications for RI and WCB, respectively.

Table 8: Sensitivity to Serial Correlation

ρ	Exact t		HC3		TWFE cluster	
	Size	Coverage	Size	Coverage	Size	Coverage
0.0	0.053	0.941	0.141	0.847	0.244	0.746
0.5	0.053	0.952	0.152	0.864	0.262	0.756
0.9	0.045	0.954	0.136	0.859	0.251	0.769

Note: $N_0 = 20$, $N_1 = 2$, $T_0 = T_1 = 10$, and errors are normal. Based on 2,000 Monte Carlo replications.

Table 9: Sensitivity to Panel Length

$T_0 = T_1$	Exact t		HC3		TWFE cluster	
	Size	Coverage	Size	Coverage	Size	Coverage
3	0.050	0.953	0.139	0.857	0.212	0.784
10	0.042	0.954	0.145	0.850	0.248	0.743
20	0.057	0.955	0.143	0.866	0.241	0.767

Note: $N_0 = 20$, $N_1 = 2$, $\rho = 0.75$, and errors are normal. Based on 2,000 Monte Carlo replications.

Table 10: Sensitivity to Non-Normal Errors

$T_0 = T_1$	Error distribution	Exact t		HC3		TWFE cluster		RI
		Size	Coverage	Size	Coverage	Size	Coverage	Size
3	Normal	0.051	0.942	0.145	0.843	0.224	0.785	0.046
3	Student- t_5	0.048	0.958	0.129	0.860	0.206	0.789	0.044
3	χ^2_2	0.056	0.947	0.137	0.867	0.201	0.799	0.049
20	Normal	0.055	0.952	0.145	0.849	0.247	0.737	0.052
20	Student- t_5	0.055	0.948	0.138	0.865	0.235	0.758	0.051
20	χ^2_2	0.063	0.937	0.146	0.870	0.250	0.758	0.059

Note: $N_0 = 20$, $N_1 = 2$, and $\rho = 0.75$. RI reports rejection probabilities only. Results are based on 2,000 Monte Carlo replications, with 999 random reassignments for RI in each replication.

Table 11: Sensitivity to Heteroskedasticity

Design	Exact t		HC3		TWFE cluster	
	Size	Coverage	Size	Coverage	Size	Coverage
Homoskedastic	0.053	0.957	0.065	0.952	0.089	0.928
Heteroskedastic	0.055	0.946	0.036	0.964	0.057	0.950

Note: $N_0 = 20$, $N_1 = 5$, $T_0 = T_1 = 10$, $\rho = 0.75$, and errors are normal. Based on 2,000 Monte Carlo replications.